



Water conservation and circular water reuse in clinical haemodialysis facilities: a scoping review

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Abstract

Introduction Haemodialysis is a life-sustaining but resource-intensive therapy with a substantial water footprint, estimated globally at 270 billion litres annually and approximately 78,000 L per patient. Green dialysis strategies increasingly focus on reducing dialysate-related water demand and recovering reverse osmosis RW; however, evidence remains dispersed across clinical studies, service reports and feasibility analyses. This scoping review mapped and synthesised literature on water conservation and reuse strategies in haemodialysis.

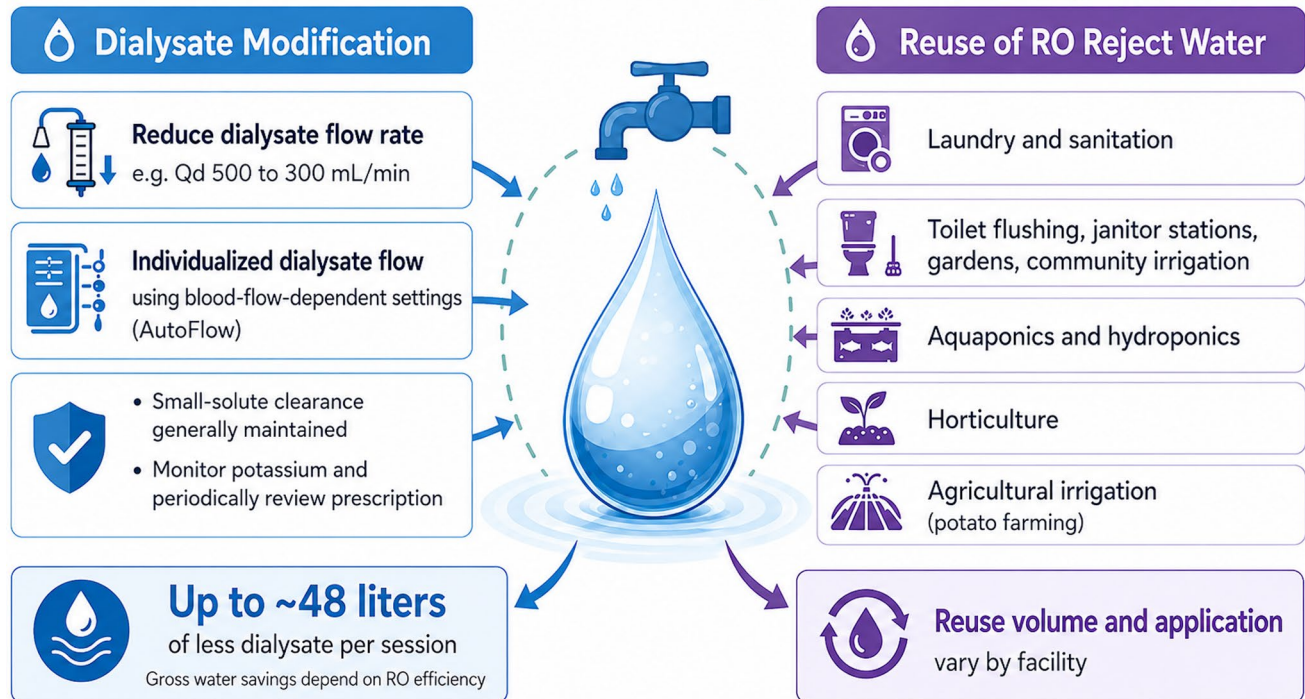
Methods A scoping review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines and registered with the Open Science Framework. PubMed, EMBASE, Web of Science, Scopus and CINAHL were systematically searched. Extracted data included study design, setting, participants or dialysis service characteristics, water use measures, reported water savings, safety, and environmental or operational outcomes.

Results 14 studies were included: nine dialysate flow modification studies and five RW reuse studies. Reported water savings ranged from 24 L/haemodialysis session to 162.3 L/treatment. Reducing dialysate flow from 500 to 300 mL/min saved approximately 48 L/session without major short-term compromise in dialysis adequacy. Reusable volumes ranged from >250 L/session to 1.1 million m³/year nationally. Facility-level reuse ranged from approximately 1.2–4.5 million L/year per centre, with applications including sanitation, autoclave steam generation, horticulture, aquaponics and agricultural irrigation.

Conclusion Water conservation in haemodialysis appears feasible through reduced dialysate flow and reject water reuse, although the magnitude of gross mains water savings depends on reverse osmosis system efficiency and local infrastructure. Evidence is primarily single centre and heterogeneous; further standardised reporting and long-term clinical, operational, environmental and implementation studies are needed.

Graphical abstract

Water Conservation Strategies in Hemodialysis



Keywords Water · Haemodialysis · Reject water · Dialysate · Sustainability

Introduction

Water scarcity, defined by the United Nations as a lack of sufficient water resources to meet the basic needs of people, including drinking, sanitation, hygiene and food production, is a critical global issue affecting billions of people (United Nations Children's Fund [47]; United Nations Water [51]). According to the World Health Organisation (WHO), around 2.2 billion people lack access to safe drinking water and 4.2 billion people lack access to safely managed sanitation services [29]. Water scarcity is not only a problem in arid and semi-arid regions, but in areas with high population densities, inadequate infrastructure and poor water management practices (United Nations Water [51]). In addition to increasing water demand associated with population growth, urbanisation and economic development, climate change is exacerbating the global water shortages (United Nations Water [50]). With only four years remaining until the 2030 deadline for the United Nations' Sustainable Development Goals (SDGs), there is an urgent need to develop practical and cost-effective water conservation approaches [48, 49].

Haemodialysis is a water-intensive medical treatment that requires substantial volumes of high-quality water [1, 2, 7]. The global burden of kidney failure requiring renal

replacement therapy continues to rise, with an estimated 4.59 million people living with end-stage kidney disease in 2023, of whom approximately 3.57 million (78%) received dialysis [34]. Haemodialysis remains the predominant dialysis modality, accounting for approximately 89% of dialysis treatments. The approximately 600 million annual global haemodialysis sessions therefore have a substantial resource footprint, particularly through water consumption, the generation of reverse osmosis (RO) reject water (RW) and spent dialysis effluent [9, 52].

Haemodialysis requires continuous circulation of dialysis fluid to facilitate the removal of solutes from blood. A conventional four-hour haemodialysis session delivered at a dialysate flow rate (Qd) of 500 mL/min requires approximately 120 L of ultrapure water for dialysate production [38, 42]. Gross mains-water consumption is higher because feed water (the incoming source water supplied to the RO system) is also used or discharged during reverse osmosis (RO) purification, pretreatment, priming, rinsing and disinfection [2, 14]. Historical reports described total consumption approaching 500 L per session, but these estimates were derived from older RO systems and should not be assumed to represent contemporary equipment. Modern system performance varies substantially by RO design, recovery setting,

feed water quality and operating conditions; contemporary measurements have reported lower per treatment water use in some centres [8]. The annual water requirement for haemodialysis has been estimated at approximately 78,000 L per patient, corresponding to a global annual requirement of around 270 billion L, although such estimates are sensitive to local equipment and treatment practices [52].

Traditionally, much of the RW generated by haemodialysis RO systems has been discharged to sewers despite often retaining a quality suitable for selected non-clinical uses. Earlier literature has sometimes described rejection using the ratio of reject flow to product flow rather than the proportion of feed water rejected. For example, production of 1 L of ultrapure water from 1.5 to 1.7 L of feed water corresponds to approximately 33–41% of feed water being rejected, not 60–70% [52]. Contemporary RO recovery and rejection rates vary by system, newer designs can reduce reject fractions substantially, whilst total facility water use also includes pretreatment and disinfection losses [8, 18]. In some haemodialysis settings, RW is recovered and repurposed for non-clinical applications such as floor cleaning, toilet flushing, gardening or farming [33, 45, 46]. More efficient RO systems may reduce water demand further, although capital cost, feed water quality, membrane fouling risk and infrastructure requirements influence implementation [1, 14, 33].

Agar et al., showed that practical conservation measures reported the recovery and reuse of >100,000 L of RW per week across facility and community applications [3]. Furthermore, Barraclough et al. found that in Victoria, Australia, only 25% of surveyed dialysis facilities were recycling RO RW, indicating that sustainable practices remain limited [7]. Therefore, development and implementation of water conservation, recovery and wastewater reuse strategies should be regarded as essential components of sustainable haemodialysis practices [9].

Previous reviews have considered selected water-saving strategies in haemodialysis [26], but a current scoping review specifically focussed on current water-conservation methods in clinical haemodialysis settings has not been identified. Preliminary findings from this work have previously been presented in conference abstract form, reflecting early recognition of the importance of water conservation in haemodialysis and reinforcing the need for a more detailed synthesis of the existing literature [40].

This scoping review aims to examine and characterise the current water conservation methods utilised in clinical haemodialysis settings, including strategies to reduce water consumption, reuse RW and recycle dialysis-related effluents as well as to identify reported implementation factors, outcomes and research gaps.

Methods

Protocol development

This scoping review follows the Preferred Reporting Items for Systematic review and Meta-Analysis (PRISMA-2020) reporting guidelines [30] with a prospectively developed protocol published in the Open Science Framework database (<https://doi.org/10.17605/OSF.IO/YRDNJ>). The scoping review addressed the following question: How can water-intensive healthcare haemodialysis reduce potable water demand and safely implement circular water reuse practices?

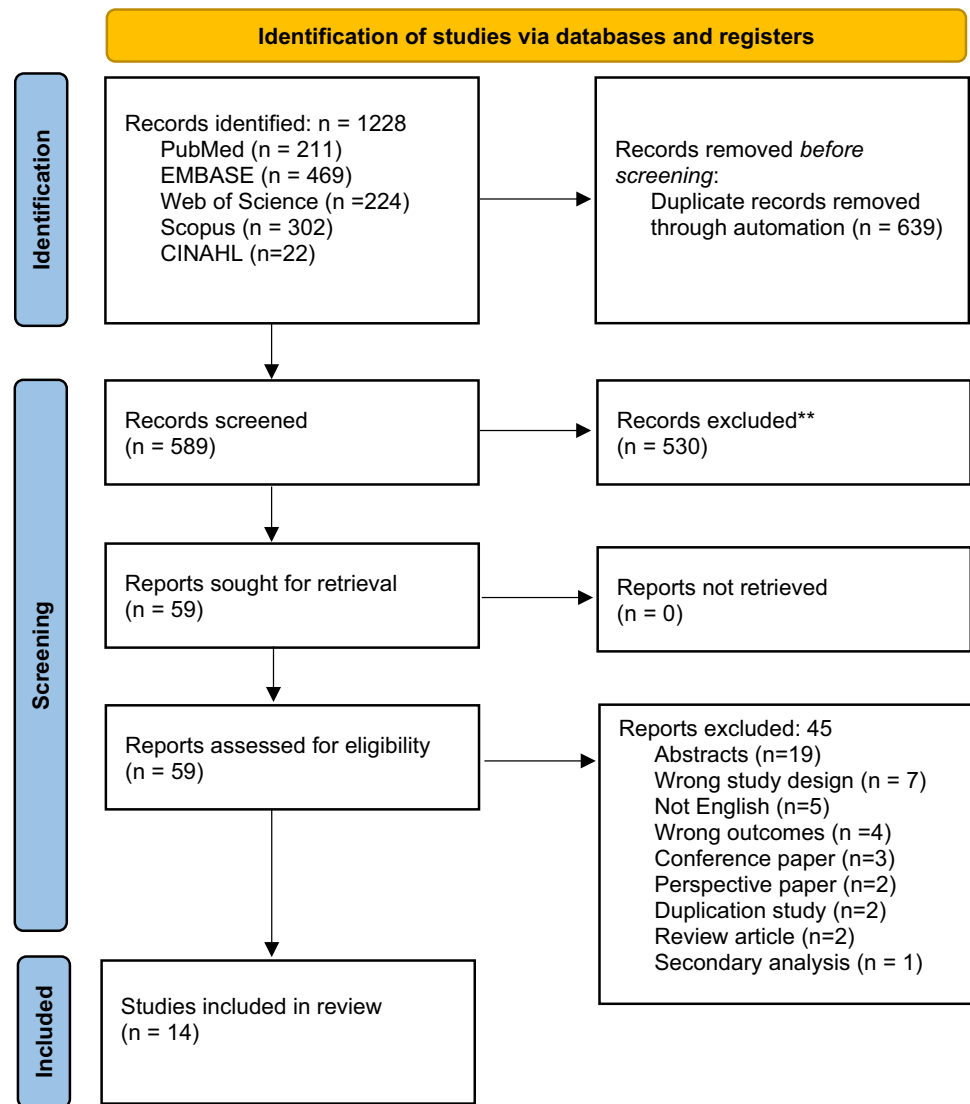
Eligibility criteria

Eligibility criteria were established using the PCC (Population, Concept, Context) framework to ensure alignment with the review aim. The population was defined as clinical haemodialysis services/facilities and where patient-level outcomes were evaluated, patients receiving facility-based haemodialysis. This broader definition reflects that the unit of analysis differed between clinical dialysate flow studies and facility-level RW implementation studies. The concept was defined as water conservation methods in clinical haemodialysis, including dialysate-flow modification and RO RW recovery or reuse. The context encompassed inpatient and outpatient clinical haemodialysis settings worldwide. Studies were excluded if they did not align with the PCC framework, discussed water use or the environmental impact of haemodialysis without addressing water conservation methods, were opinion pieces, commentaries, editorials, letters, perspectives, protocols, abstracts or conference presentations, or were not available in English. No date limit was applied.

Search strategy

Six authors (AS, KB, NM, TT, AG and MO) developed the search strategy to align with the review question and eligibility criteria. A combination of a MeSH term ('Waste-water') and free-text title/abstract terms was combined using Boolean operators to systematically search PubMed, EMBASE, Web of Science, CINAHL and Scopus for studies published from inception to 7 March 2026. Preliminary searches were first conducted in PubMed and EMBASE and subsequently translated for use in Web of Science, CINAHL and Scopus utilising Polyglot, accessed via the Evidence Review Accelerator (TERA) [13]. Whilst the search syntax was translated across databases using Polyglot, the mapping of controlled vocabulary terms was checked and adapted manually. In addition, the reference lists of relevant systematic reviews were hand-searched to identify any further eligible studies.

Fig. 1 PRISMA flow diagram for study selection process



Screening and selection process

Identified articles were independently screened by title and abstract against the eligibility criteria within Covidence. Following retrieval of the full-text articles, authors screened each full text for inclusion. Discrepancies were resolved by consensus.

Data charting process

Data were charted using a customised extraction template to ensure consistent capture of study characteristics, intervention details and water-related outcomes relevant to the review question. The draft data extraction template was reviewed, piloted and refined prior to data extraction of included articles being undertaken. As the included studies clustered into two main categories, the extraction template was organised separately for dialysate flow rate modification and RO RW

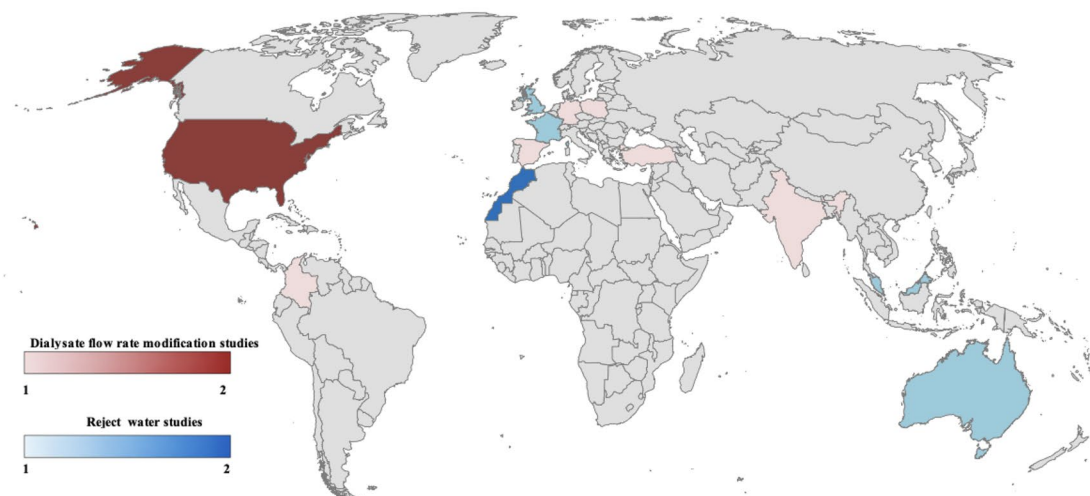
reuse studies. This allowed study characteristics and outcome data to be charted consistently according to intervention type, whilst maintaining a common framework across all included studies. Extracted variables included author and year, study design, setting, intervention characteristics, and quantitative water-related outcomes relevant to each category.

Synthesis of dialysate flow rate modification and RW studies

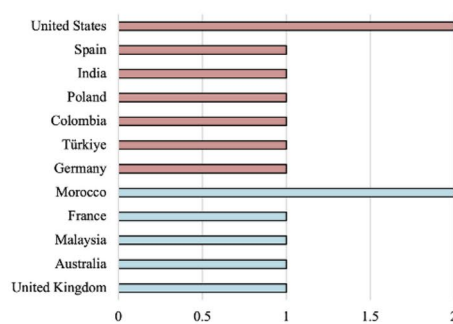
Given the heterogeneity in intervention types, outcome definitions and reporting units across included studies, a descriptive quantitative synthesis was undertaken. For dialysate flow rate modification studies, extracted outcomes included dialysate or water use per haemodialysis session, water or dialysate saved per session and total reported savings over the study period. Only facility-based

A.

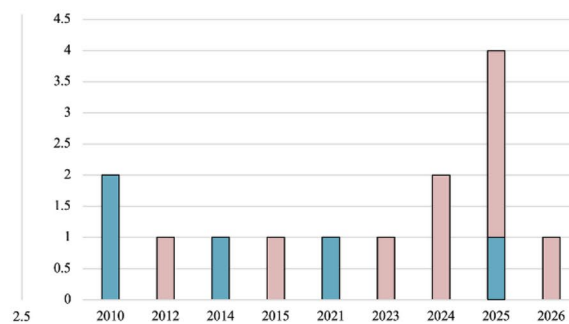
World map of included studies



B. Number of publications per country



C. Year of publication



D.

Reject water reuse and dialysate flow rate modification

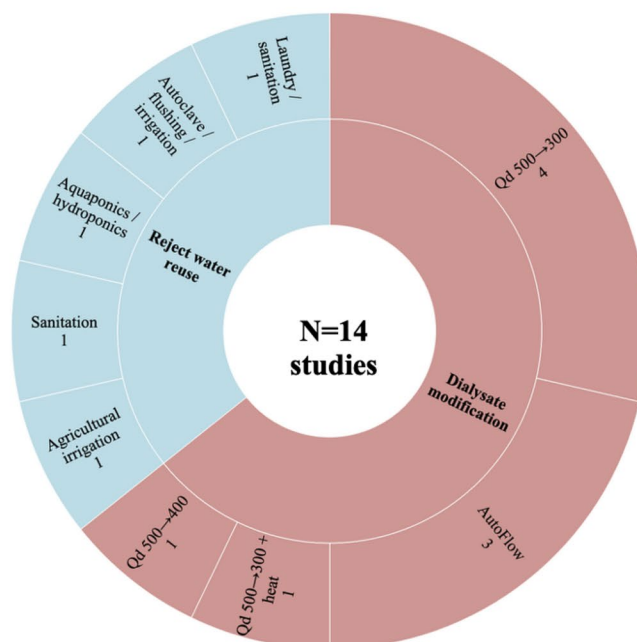


Fig. 2 Key characteristics of included studies

haemodialysis data were included to support comparability across studies. Where studies reported both dialysate and purified water volumes, these were recorded separately and were not combined. Because most included studies did not report RO recovery, we have not retrospectively converted dialysate savings into gross mains-water savings. For studies evaluating RO RW generation and reuse, extracted outcomes included total water use per session, RW volume generated and RW volume reused or saved. When RW reuse was reported only over daily or weekly periods, annual estimates were calculated using 365-day or 52-week standardisation and were clearly identified as inferred values. Where study terminology varied, “treatment” and “session” were classified the same to maintain consistency. Results were therefore synthesised narratively and presented in tabular format.

Results

Study selection

The study selection process is summarised in the PRISMA flow diagram presented in Fig. 1. The systematic search identified 1228 records, of which 639 duplicates were removed. The remaining records were screened by title and abstract, after which 530 were excluded. This resulted in 59 articles being assessed in full text against the eligibility criteria. Of these, 45 were excluded with almost perfect agreement (Cohen’s kappa = 0.932) [24], leaving 14 studies included in the final review (Fig. 1). One study reported only theoretical water savings without providing extractable quantitative data [15].

Characteristics of included studies

The key characteristics of included studies are described in Fig. 2. The 14 articles represented findings from 12 countries, with two studies from the United States [15, 31] and two from Morocco [5, 44]. Nine studies focussed on dialysate flow modification [4–6, 15, 16, 31, 35, 38, 42] and five on RW reuse [2, 12, 14, 33, 44]. Dialysate flow studies included fixed lower Qd prescriptions (for example, 300 or 400 mL/min) and individualised blood-flow-dependent strategies. These were treated as distinct approaches: fixed lower Qd is a prescribed constant flow, whereas variable flow strategies such as AutoFlow (AF) adjust Qd in relation to blood flow rate (Qb), defined as the rate of blood pumped through the extracorporeal circuit. Reject water studies were predominantly facility-level implementation or feasibility reports from the United Kingdom, Australia, Malaysia, France and Morocco and evaluated infrastructure or reuse pathways rather than patient-level outcomes.

Dialysate flow rate modification studies

Nine studies investigating dialysate flow modification strategies in clinical haemodialysis were included (Table 1). Eight of nine studies reported quantitative data on dialysate flow or water use at the sessional level [4–6, 16, 31, 35, 38, 42] whilst one study provided only theoretical estimates [15]. Studies were mostly conducted as observational studies, including prospective cross-over studies, prospective non-randomised trials, retrospective observational studies and post hoc cohort analyses. Studies published in 2012 and 2015 were mainly single-centre prospective cross-over designs [5, 6]. Later studies (2023–2026) broadened methodologically to include multicentre observational work [31] and practice-change analysis [16], as well as retrospective [38] and post hoc analysis [35]. Across studies, baseline dialysate consumption for a standard four-hour haemodialysis session at Qd 500 mL/min ranged from approximately 120–150 L per session. Values of 120 L/session were reported by Albalade et al. [6], Alayoud et al. [5], Ramos et al. [35], and Solomon et al. [42]. Two studies reported similar baseline dialysis consumption of 127.5 L/session [16] and 126 L/session [31], whereas Aktaş et al. reported an average of 150 L/session at Qd 500 mL/min [4].

Sessional dialysate or purified water savings varied according to the intervention. Reducing fixed Qd from 500 to 300 mL/min was associated with approximately 48 L less dialysate per four-hour session in three independent studies [4, 38, 42]. Reducing Qd from 500 to 400 mL/min saved approximately 24 L/session [6, 35]. Individualised blood-flow-dependent strategies also reduced dialysate use, with David et al. reporting a 35.7 L/session reduction [16]. The largest reported reduction was in a multicentre comparison of green haemodialysis versus conventional systems, where total treatment water use was reduced by 162.3 L/treatment [31]. Importantly, most fixed Qd studies quantified dialysate or ultrapure water savings rather than gross mains water savings. The corresponding reduction in feed water demand depends on RO recovery efficiency and therefore cannot be validly converted to a single gross water value when system specific recovery data are not reported. No dialysate flow rate modification studies reported details on their specific RO models or efficiency.

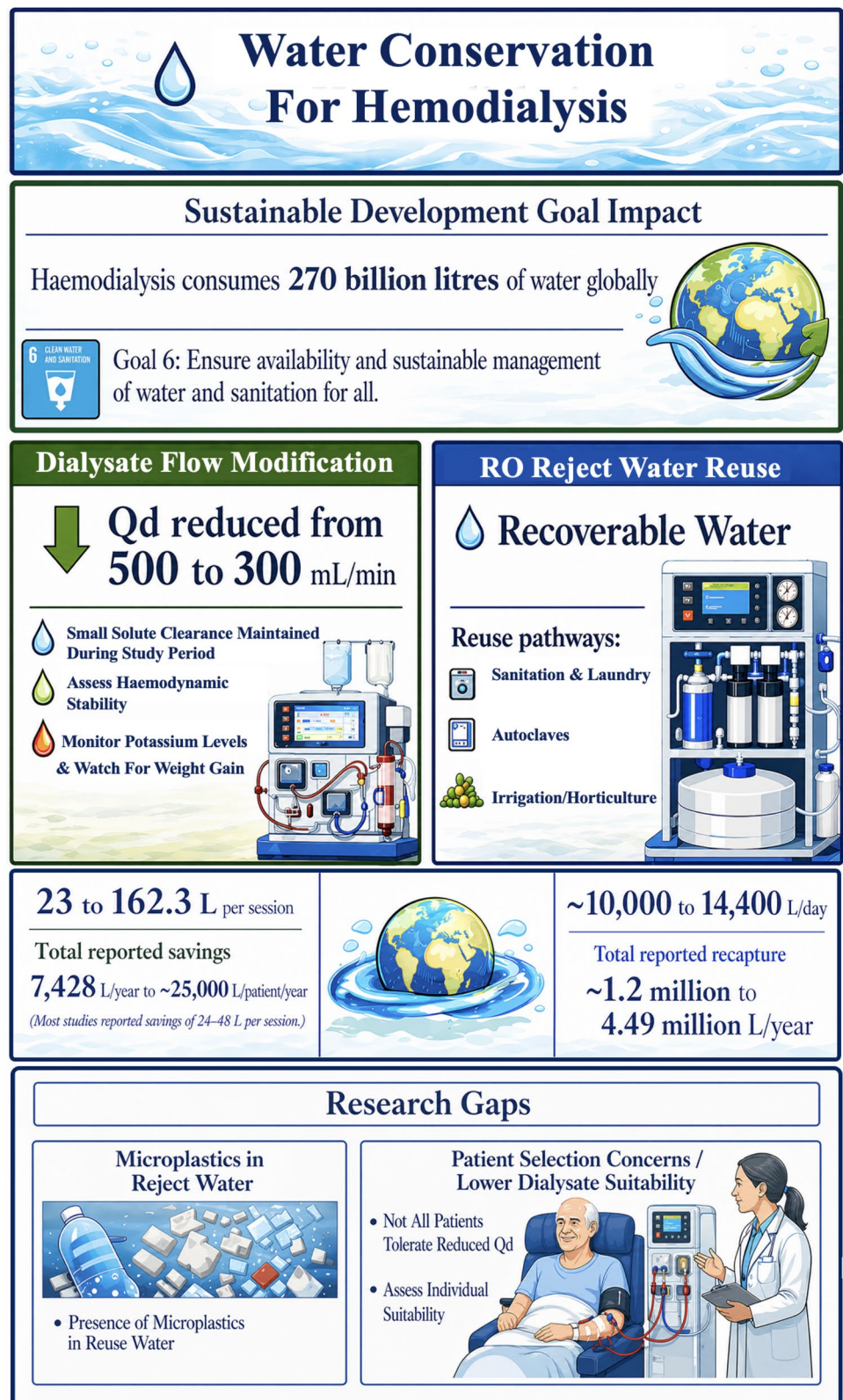
Where reported, total water savings varied substantially across studies, influenced by study duration, patient numbers and reporting units. Reported totals included 13,824 L over 12 weeks, ~7,500 L per patient per year [4], 19,656 versus 44,979.5 L/year [31], 194,688 L/year in a 26-patient unit [38] and 1,440 L/day purified water saved at unit level [42]. Overall, despite heterogeneity in reporting, all studies demonstrated that reducing dialysate flow rate or individualising dialysate delivery was associated with measurable reductions in dialysate or water consumption, whilst dialysis adequacy and small solute clearance was generally maintained Fig. 3.

Table 1 Water use and water savings in dialysate flow rate modification studies ($n=9$)

Study	Study design	Intervention	Dialysis machine/model	Dialyzer/membrane	Sessional dialysate or Water use	Water/Dialysate saved per session	*Total savings reported
[6]	Prospective single-centre crossover study	Fixed Qd 400 mL/min vs Fixed Qd 500 mL/min vs Fixed Qd 700 mL/min plus a separate calculated AutoFlow (AF) arm	AK200 ($n=21$); Fresenius 5008 ($n=8$); Artis ($n=2$)	Xenium H ($n=9$); Xenium M ($n=8$); FX80 ($n=7$); Polyflux 210 ($n=7$)	96 L (Qd400 mL/min) 120 L (Qd500 mL/min) 168 L (Qd700 mL/min) AF: not directly reported; mean achieved Qd 507 (52.6) mL/min	24 L/session 48 L/session 72 L/session AF: no sessional saving reported	7428 L/year at (Qd 500 mL/min vs. Qd 700 mL/min); 11,232 L/year (Qd 400 mL/min vs. Qd 700 mL/min); AF: not reported
[5]	Prospective single-centre crossover study	Automated AutoFlow (AF) vs fixed Qd 500 mL/min	Fresenius 5008 Therapy System	Polyflux H140, H170 and H210 (Polyamix)	96 L (AF) 120 L (Qd500 mL/min) 89 L of PW (AF) 112 L of PW (Qd500 mL/min)	24 L/session 23 L PW/session	Not reported
[4]	Prospective multicentre study	Fixed Qd 300 mL/min vs Fixed Qd 500 mL/min	Fresenius 4008 or B. Braun Dialogue+	Fresenius Medical Care, Baxter and Clearum Medtronic dialyzers (exact models not reported)	102 L/session (Qd 300 mL/min) vs 150 L/session (Qd500 mL/min)	48 L/session	~7500 L/year/patient
[15]	Prospective non-randomised trial	Fixed Qd 300 mL/min vs ≥ 500 mL/min	Fresenius FMC 2008 K	Exact dialyzer model not reported	Not reported	Not reported	Not reported
[16]	Observational change-of-practice study	Fixed Qd 500 mL/min vs individualised blood-flow-dependent Qd (Qd = $1.2 \times Q_b$)	Not reported	Not reported	127.5 L/session (Qd 500 mL/min) vs 92 L/session (AF)	35.7 L/session	Not reported
[31]	Retrospective multicentre observational study	Fixed Qd 300 mL/min (Green HD) vs Fixed Qd 500 mL/min	Green HD: Tablo® HD System (Outset Medical), internal dual RO; conventional machine models not specified	High-flux dialyzer (exact model not reported)	126.0 L/session (Qd 300 mL/min) vs 288.3 L/session (Qd 500 mL/min)	162.3 L/session	19,656 vs. 44,979.5 L/year; ~25,000 L saved/person/year
[35]	Post hoc retrospective cohort analysis	Fixed Qd 400 mL/min vs Fixed Qd 500 mL/min (HDx)	Machine model not reported	Theranova® 400 membrane (Baxter)	96 L/session (Qd 400 mL/min) vs 120 L/session (Qd 500 mL/min)	24 L/session	13,824 L over 12 weeks
[38]	Retrospective observational study	Fixed Qd 300 mL/min vs Fixed 500 mL/min	Not reported	Not reported	72 L/session (Qd 300 mL/min) vs ≥ 120 L/session (Qd500 mL/min)	≥ 48 L/session	194,688 L/year (26-patient unit)
[42]	Single-centre non-randomized open-label crossover pilot trial	Fixed Qd 500 mL/min (NTNF) vs Fixed Qd 500 mL/min (HTNF) vs Fixed Qd 300 (HTNF)	Fresenius 4008S	Fresenius F6HPS polysulfone, 1.3 m ² (KUF 13 mL/h/mmHg; K0A 731 mL/min)	120 L/session (NTNF / HTNF at Qd 500 mL/min) vs 72 L/session (HTLF at Qd 300 mL/min)	48 L/session (40% reduction)	1440 L/day of PW saved (unit-level)

HD haemodialysis, Qd Dialysate flow rate, HDx expanded haemodialysis (dialysers with membranes capable of enhanced clearance of medium-sized molecules due to expanded pore capacity), AF autoflow, PW purified water, NTNF normal-temperature normal-flow (37 °C), HTNF high-temperature normal-flow (38.5 °C). *Reported total sessional savings are retained in the units used by the source study. Dialysate or purified-water savings should not be interpreted as gross mains-water savings unless RO recovery/feed-water data were reported

Fig. 3 Overview of water conservation strategies identified in haemodialysis. The infographic summarises the two principal approaches reported in the included studies: reducing dialysate-related water use through dialysate flow rate modification and repurposing reverse osmosis RW for secondary applications



Reject water recapturing studies

Five studies reporting on RO RW generation and reuse in clinical haemodialysis settings are included (Table 2). Reject water studies were predominately service-level implementation reports, technical case studies, or feasibility/modelling studies, with a strong focus on infrastructure, reuse pathways and operational outcomes. These are reflective of RW reuse as largely a health care system, engineering and implementation challenge rather than a patient-level prescription issue. Four different studies by Agar et al., Connor et al., Ponson et al. and Chang et al. primarily described implementation of reuse systems and volumes of RW that be recovered or redirected [2, 12, 14, 33], whilst Tarrass et al. extended this to a national-scale feasibility model [44].

Across five studies, substantial volumes of RW were identified as available for reuse. Connor et al., reported >250 L of RW per session, with reuse of 14,400 L/day and 4.5 million L/year at a single facility [14]. Agar et al. reported ~65,000 L/week of facility-based RW reuse, corresponding to ~3.38 million L/year [2]. Chang et al., reported 10,000–12,000 L/day of RW redirected to aquaponics and horticulture systems, equating to an estimated average of ~4.0 million L/year [12]. Ponson et al. reported 2834 m³/year of RW, of which 1200 m³/year was considered appropriate for sanitation reuse [33]. Tarrass et al. estimated national RW generation in Morocco at ~1.1 million m³/year although this represented modelled production rather than direct implementation data [44]. Two studies directly reported total sessional water consumption for in-centre haemodialysis, with values consistently of approximately 493–500 L per treatment. Connor et al., reported approximately 400 L/session for haemodialysis-grade water production and approximately 500 L/session when priming and rinsing were included [14], whilst Agar et al., reported 493 L/treatment for in-centre dialysis [2]. No other studies provided directly comparable sessional water use data. Overall, facility-level water reuse ranged from approximately 1.2–4.5 million L/year/centre, with higher values associated with large or optimised systems. Direct comparisons between studies were limited by differences in reporting units (sessional, daily, weekly, annual), reuse endpoints and whether values represented theoretical availability or actual reuse.

Discussion

This review demonstrates that water conservation in haemodialysis is achievable and necessary, showing that meaningful improvements can be achieved in two main ways: dialysate flow rate modification and repurposing of RO RW.

Table 2 Water use and water savings across RW studies (*n*=5)

Study	Study design	RO/water treatment system	Reuse infrastructure/technical details	Total water use (per session)	Reject water generated	Water reused or saved	Timeframe	Water reuse applications
[14]	Implementation case study	RO model not reported; historical system reported to reject ~two-thirds of feed water	Reject water gravity drains to basement recovery tank, then pumped to rooftop grey water tank; float switch control and diverter valves; salvage pump 2,400 L/h. New build integration included tank/control panel and pipework	~400 L/session (HD-grade water); ~500 L/session total including priming/rinsing	>250 L/session	14,400 L/day; 4,492,000 L/year	Daily and annual	Laundry and sanitation

Table 2 (continued)

Study	Study design	RO/water treatment system	Reuse infrastructure/technical details	Total water use (per session)	Reject water generated	Water reused or saved	Timeframe	Water reuse applications
Agar et al., 2010	Service-level report	In-centre: Renoflow 5 single RO (installed 1998); Satellite 1: Renoflow 6 double RO (2001); Satellite 2: Renoflow 2 single RO (2001)	In-centre: 2,500 L holding tank, commercial pump and 2 × 35,000 L roof storage tanks; satellite water distributed by tanker/water cart. RO generation rates: 15, 18 and 8 L/min across the three facilities	493 L/treatment (in-centre HD only)	Not reported	~65,000 L/week (facility-based reuse only) ~3.38 million L/year*	Weekly and annual	Autoclave steam generation, toilet flushing, janitor stations, garden maintenance, sporting fields, schools, aged-care gardens
[12]	Implementation report	Direct-feed 4-membrane RO system; nominal flow ~20 L/min; recovery adjustable and set at 60%; closed-loop distribution	Reject water line re-piped and pumped to fish tanks; water then distributed to aquaponics/hydroponics and horticulture; overflow valve diverts excess water	Not reported per session	10,000–12,000 L/day (RW)	10,000–12,000 L/day reused (assumed full reuse in system) ~3.65–4.38 million L/year*	Daily and annual	Aquaponics, hydroponics and horticulture
[33]	Technical economic case study	Fresenius water treatment with common pretreatment and 2 AquaB DUO biosmosis units; measured water conversion factor 69% ± 3%	Proposed sanitation system: ~200 m PVC pipework, lift pump, 20 control valves and 5m ³ tank; contingency mains water inlet; targeted 20 toilets	Not reported per session; total 9286.78 m ³ /year processed	2834 m ³ /year RW	1200 m ³ /year reused (feasible sanitation reuse)	Annual	Building sanitation, particularly toilets
[44]	Feasibility and modelling study	No single RO make/model: national-scale modelling across Moroccan facilities	Modelled agricultural reuse pathway; no single facility capture/storage system evaluated	Not reported per session	~1.1 million m ³ /year (national estimate)	Not directly reported (modelled agricultural reuse)	Annual	Agricultural irrigation, specifically potato farming

RO reverse osmosis. *Inferred values derived from reported daily or weekly values using standard calendar conversions

In the dialysate flow rate modification literature, several studies reported that lowering dialysate flow can reduce water or dialysate consumption without clinically significant short-term compromise in adequacy. Albalade et al. found that increasing Qd above 400 mL/min provided only limited gains in dialysis adequacy, whilst substantially increasing dialysate consumption and concluded that extending dialysis time was a more efficient trade-off than using higher water volumes [6]. Similarly, Solomon et al. reported that reducing Qd from 500 to 300 mL/min lowered purified water use by 40% without compromising adequacy or short-term safety in young, haemodynamically stable patients [42].

Variable flow and fixed low-Qd prescriptions should be distinguished. Fixed-flow strategies prescribe a constant Qd, whereas variable flow, such as AF, is a machine-dependent function that dynamically links Qd to effective blood flow (Qb), typically using a Qd:Qb factor. In Alayoud et al., automated AF was delivered using the Fresenius 5008 Therapy System with Polyflux H 140, 170 and 210 dialyzers; treatment time, Qb and dialyzer type/size were held constant [5]. Albalade et al. compared separate fixed Qd arms (400, 500 and 700 mL/min) with a calculated AF arm across AK200, Fresenius 5008 and Artis monitors using Xenium H, Xenium M, FX80 and Polyflux 210 dialysers, therefore the AF condition should not be interpreted as equivalent to the fixed 400 mL/min arm [6]. David et al. individualised Qd so that it remained 20% above blood flow during online HDF [16]. These technical differences are important because the availability and implementation of automated Qd adjustment depend on machine platform, software and consumables [18]. Across the included studies, individualised flow strategies reduced dialysate use without clear short-term compromise in adequacy, but prescription suitability should be reviewed periodically as patient blood flow, access function, biochemical control and clinical status change. An additional, yet to be determined, opportunity may be to combine AutoFlow with intradialytic cycling as exercise during haemodialysis has been proposed to enhance uraemic toxin clearance, stabilise vascular resistance, improve blood pressure stability and treatment tolerance without increasing symptomatic intradialytic hypotension and other adverse events [22, 23, 37]. Therefore, it is plausible that variable flow combined with intradialytic cycling exercise could be an effective tool in selected patients, potentially preserving adequacy and preventing physical decline whilst maximising dialysate and water savings.

In parallel, RW studies illustrated that large volumes of otherwise discharged water can be redirected to useful secondary applications. The included literature demonstrated implementation across a range of economic and healthcare settings, although the most developed examples of fixed-site capture and redistribution infrastructure were reported

from high-income countries, including the United Kingdom, Australia and France, where RW was reused for applications such as sanitation, laundry, sterilisation and irrigation. This pattern in the published evidence should not, however, be interpreted as indicating that national income classification determines the most appropriate reuse strategy. Feasibility is primarily site-specific and depends on factors including RO plant configuration, access to the RW line, available space for storage, distance to the reuse endpoint, plumbing and cross-connection controls, water quality, infection-prevention requirements, regulatory approval, capital cost and the presence of a practical local demand for the recovered water. Facilities with greater capital resources and established water-reuse infrastructure may be better positioned to implement more complex capture and redistribution systems, whereas lower-cost or context-adapted reuse pathways may provide more immediately achievable options where infrastructure or capital is constrained.

Connor et al. reported 4.49 million L/year of recovered RW with net annual financial savings exceeding £10,000/year [14], whilst Ponson et al. estimated a 5.8-year return on investment for sanitation reuse of 1200 m³/year [33]. Studies from settings such as Malaysia and Morocco further demonstrate that alternative applications, including aquaponics, horticulture and agricultural reuse, may also be feasible where facility localisation, surrounding land use, infrastructure and local regulation support these pathways. Agricultural reuse, for example, may be appropriate for some facilities but impractical for urban dialysis centres without access to suitable land or distribution infrastructure. The appropriate reuse strategy should therefore be determined at the facility level rather than inferred from country income classification alone.

Existing dialysis facilities may be able to retrofit RW capture and redistribution systems without replacing the RO plant where the reject stream can be safely intercepted and routed. However, the feasibility and cost of retrofitting will depend on the existing plumbing configuration, available space, storage requirements and proximity of suitable reuse endpoints. Major refurbishment, scheduled RO replacement and particularly the construction of new dialysis units provide favourable opportunities to incorporate storage, pipework, monitoring and fit-for-purpose reuse infrastructure into facility design from the outset. Given the anticipated growth in the global dialysis population and the associated increase in water demand, assessment of RW capture and appropriate reuse should therefore be considered during the planning of new dialysis facilities across healthcare settings, with the final reuse pathway selected according to local technical, regulatory, economic and environmental conditions.

The practical and safety considerations differ between the two water sustainability approaches. Across dialysate-flow studies, there was little evidence of patient harm associated

with reducing or individualising dialysate flow, although important cautions emerged. Rydzewska-Rosołowska et al. found that reducing Qd from 500 to 300 mL/min did not significantly worsen urea reduction ratio or dialysis adequacy but was associated with a small statistically significant increase in mean pre-dialysis potassium [38]. Solomon et al. reported preserved adequacy with a high-temperature, low-flow prescription but observed one episode of symptomatic hypotension requiring dialysis termination, because both temperature and flow were altered, the event cannot be attributed to low flow alone [42]. Coritsidis et al. also suggested that longer exposure to Qd 300 mL/min may be associated with increased serum potassium in a small number of patients [15]. Accordingly, lower-flow prescriptions should be individualised and periodically reviewed, particularly when vascular access, Qb, interdialytic weight gain, treatment time or biochemical control changes. A 48 L reduction in dialysate production also reduces the amount of RO permeate that must be produced and therefore reduces gross feed water demand and reject generation, however, the magnitude of that upstream saving is RO-specific. For example, at 60–75% RO recovery, 48 L less product water would correspond to approximately 64–80 L less feed water, whereas older low recovery systems could require considerably more.

In contrast, the main challenge with RW reuse is less about immediate patient adequacy and instead demonstrating that the water is chemically and biologically suitable for secondary use. Across the RW studies, reuse was generally complimented by water quality assessment, although the rigour and scope of testing varied. Agar et al., provided the strongest quality assessment through repeated water testing to demonstrate that RO RW was essentially comparable with local potable water, apart from minor conductivity differences within accepted limits for reuse [2]. Chang et al. similarly performed pre-implementation water testing and reported parameters suitable for aquaculture, supporting reuse in aquaponics, hydroponics, and horticulture [12]. Ponson et al. used physicochemical testing, including sodium measurement, to justify restricting reuse to sanitation applications rather than broader reuse pathways [33]. Tarrass et al. concluded that RW met irrigation-water standards for agricultural reuse [44], whilst Connor et al. described RW as high-grade grey water that commonly met potable criteria [14], albeit with the implication that local verification remained necessary before implementation.

In Australia, reuse of RW is especially relevant because recycled water is managed under the Australian Guidelines for Water Recycling (AGWR), which use a fit-for-purpose, risk-based framework requiring hazard identification, risk assessment, treatment validation, and ongoing monitoring

according to the intended end use [27]. Reject water reuse should be tailored to the proposed application, such as irrigation, sanitation, horticulture and therefore may need to include not only microbiological indicators such as *E. coli* or thermotolerant coliform testing, but checking chemical and salinity parameters, such as electrical conductivity, total dissolved solids, sodium, chloride and boron, particularly when there is risk of soil degradation, plant toxicity, or ground-water impact [17].

An additional, largely unexplored consideration is whether plastic components of dialysis water-treatment and delivery systems may contribute micro- or nanoplastics (MNPs) or plastic additive chemicals to RW streams. Whilst this was not evaluated in the included studies, emerging reviews show that MNPs can accumulate in respiratory [19], cardiovascular [20] and urinary systems [28] and cause harmful cellular effects. None of the included RW studies measured MNPs, so their presence, concentration and clinical or environmental significance in dialysis RW cannot currently be quantified. Evidence from other medical-device settings demonstrates particle shedding from plastic tubing and extracorporeal circuits [10, 21, 39], whilst experimental and environmental literature indicates that small plastic particles may interact with plants and soils [55]. These findings justify investigation but do not establish MNPs as a proven contraindication to RW reuse. Future studies should prospectively sample feed water, RO product water and RW, use contamination-controlled collection with procedural blanks; report particle number, size distribution and polymer type using validated analytical methods. The analysis of MNP and plastic additive chemicals however, has been noted to be extremely challenging requiring specialised equipment and expertise [36, 41, 53].

Irrigation or other uses that may create aerosol exposure should be assessed separately from enclosed sanitation uses. At present, the absence of standardised health- or reuse-based threshold values is an important evidence gap, so MNP monitoring should be incorporated into research protocols rather than used as an unsupported reason to prohibit reuse.

A further limitation of the historical literature is the difficulty of comparing gross water use across generations of dialysis and RO equipment. Older sustainability papers frequently cite approximately 500 L per treatment and very high RO rejection fractions, whereas contemporary systems can operate at substantially higher recovery. In a 2024 real-world assessment of two Australian satellite units, Barraclough et al. reported lower per treatment water consumption than historical estimates, whilst also demonstrating that

total water not delivered as dialysis product water includes RO reject, pretreatment and disinfection requirements [8]. Contemporary modelling from two Royal Free London Kidney Care Centres similarly reported approximately 306 L of raw water use per HDF treatment, of which 144 L was used for treatment at a combined flow of 600 mL/min, leaving approximately 162 L associated with RO reject, pretreatment and disinfection processes (Steinbach [43]). This distinction is essential: membrane reject fraction, total RO-plant water loss and whole treatment mains water consumption are not interchangeable metrics. Future sustainability studies should therefore report RO make/model or recovery setting, feed water volume, product water volume, RW volume, pretreatment/disinfection use, number of treatments, treatment duration, patient-shifts and unit size. Standardised reporting would permit meaningful comparison between facilities and more reliable estimation of the true gross water benefit of reduced Qd or RW reuse.

A notable finding of this review was the disconnect between the volume of discussion surrounding sustainable dialysis and the relatively small number of studies demonstrating practically implemented water conservation strategies. In addition to the included studies, a substantial number of excluded records comprised of abstracts ($n=19$), conference papers ($n=3$) and perspective pieces ($n=2$), (Fig 1), indicating that the issue is widely recognised but less frequently translated into evaluated service-level initiatives. Considering the COM-B Theoretical Domains Framework (TDF) [11, 25], this gap likely reflects barriers in capability, opportunity and motivation. Services may lack the technical knowledge and skills required to operationalise the change such as infrastructure limitations, cost and regulatory complexity as well as remain hesitant in the absence of strong safety, economic and long-term effectiveness data [32, 54]. These findings suggest that advancing water conservation in haemodialysis will require implementation science approaches that address behavioural, organisation and system-level barriers, rather than relying on awareness alone.

Potential barriers to implementation were considered using the COM-B model and mapped to relevant TDF domains to illustrate why promising water-conservation strategies may remain under-implemented despite increasing awareness (Table 3).

Conclusions

Water conservation in haemodialysis is feasible, but its achievable magnitude is dependent on dialysis prescription, RO efficiency, facility design and local reuse opportunities. This scoping review identified two main approaches: reducing dialysate-related water consumption and repurposing

RO RW for secondary use. The 14 included studies suggest that meaningful reductions in water use can be achieved without major short-term compromise in dialysis adequacy in selected patients, whilst RW can be redirected to practical applications when supported by appropriate quality assessment and infrastructure. However, the evidence base is limited by predominantly single-centre designs, heterogeneous outcome measures, inconsistent distinction between dialysate/product water savings and gross mains water savings, incomplete reporting of RO efficiency as well as limited long-term clinical, environmental and economic evaluation. Future studies should standardise water accounting and report unit size, treatment numbers, RO configuration/recovery and all major water streams. New dialysis units and major refurbishments should routinely assess provision for RW capture and fit-for-purpose reuse at the design stage. Lower Qd prescriptions should remain individualised and periodically reviewed whilst future RO RW research should include emerging contaminants such as microplastics using standardised analytical methods. These steps would provide a more robust evidence base for reducing the water footprint of dialysis whilst maintaining safe, high-quality patient care.

Author comments/recommendations

To complement the systematic synthesis presented in this review, Table 4 summarises the key characteristics, findings and implementation considerations of the included studies.

Translational statement/lay summary

Haemodialysis is a life-saving treatment, but it uses very large amounts of water and generates substantial wastewater. This review examined published studies on practical ways to conserve water in clinical haemodialysis. Two main approaches were identified: reducing water use during dialysis by lowering or individualising dialysate flow and reusing reverse osmosis RW for non-potable purposes such as sanitisation, irrigation or horticulture. The findings suggest that meaningful water savings are achievable, often without major short-term effects on dialysis adequacy in selected patients, although monitoring remains important. Reject water reuse studies found that large volumes of water can be repurposed when supported by appropriate water quality assessment. The evidence, however, remains limited with relatively few real-world implementation studies. These findings highlight an important opportunity to reduce the environmental footprint of haemodialysis and support more sustainable renal care, whilst also identifying areas where further clinical and implementation research is needed.

Table 3 Potential implementation barriers to dialysate flow rate modification and RO RW reuse strategies in haemodialysis, interpreted using the COM-B model and TDF

COM-B domain	TDF domain	Barriers	Facilitators
Capability	Knowledge	Lack of expertise in plumbing, water-treatment systems, recycled-water governance and hospital building/modification requirements. Limited awareness of how water-saving strategies can be implemented safely in clinical dialysis settings	Increased collaboration between key stakeholders including clinicians, administrators, engineers, architects, infection-control teams, facilities staff and tradespeople. Development of practical guidance documents and implementation case studies
	Skills	Limited technical skills to modify dialysis prescriptions, integrate reuse systems, interpret recycled-water requirements, or manage upgraded infrastructure	Staff training in water-saving dialysis prescriptions, infrastructure operation and water quality management. Access to technical support
	Memory, attention, and decision processes	Sustainability initiatives may be overlooked in routine dialysis environments where immediate clinical issues are priority. Difficulty integrating water conservation into routine decision making	Use of checklists, protocols and decision-support tools (AD) to embed sustainability into routine care and infrastructure planning
	Behavioural regulation	Lack of standardised protocols for implementing reduced Qd prescriptions of RW reuse pathways. Variable monitoring and follow-up practices	Development of local protocols, audit cycles, implementation pathways and clear frameworks for patient safety and water quality assurance
Opportunity	Environmental context and resources	Legacy/heritage infrastructure, plumbing/retrofitting challenges, limited space, equipment complexity, cost of installation and maintenance, lack of access to reuse pathways	Fit-for-purpose planning, phased implementation, dedicated capital support, integration into new builds/refurbishments and use of established recycled water frameworks such as AGWR
Motivation	Social influences	Sustainability not yet embedded in renal service culture, limited leadership support, competing clinical priorities, uncertainty surrounding who is responsible for implementation	Strong leadership support, shared ownership across clinical and non-clinical teams, institutional culture that values environmental stewardship alongside patient care
	Beliefs about capabilities	Concerns about dialysis adequacy (potassium, hypotension or water quality). Uncertainty about economic return. Limited confidence in feasibility, sustainability seen as secondary to clinical delivery	Demonstration studies showing maintained adequacy, manageable safety signals and practical reuse outcomes. Sharing of successful implementation examples to build confidence
	Social or professional role identity	Perception that water conservation is not a core responsibility of renal clinicals of dialysis services. Sustainability may be viewed as outside the traditional scope of care	Framing water conservation as part of high-quality, responsible healthcare and aligning it with professional values, environmental stewardship and healthcare sustainability
	Beliefs about consequences	Concerns regarding patient safety, regulatory risk, infection risk, environmental contamination or unintended harm. Fear that modifications may compromise care of generate liability	Clear evidence that lower dialysate flow can maintain adequacy in selected patients and that RW reuse can be safely managed when aligned with fit-for-purpose water quality assessment and governance
Intentions	Emotions	Resistance to changing established dialysis routines, burnout due to high work demands and anxiety about introducing unfamiliar sustainability practices. Limited incentives or reinforcement for adopting change	Leadership endorsement, viable institutional support, recognition of sustainability efforts and framing change as achievable rather than burdensome
	Goals	Sustainability goals may be poorly defined or not prioritised relative to service throughput, staffing and immediate clinical outcomes	Inclusion of water conservation targets in departmental quality improvement plan, sustainability strategies and renal service performance indicators
	Intentions	General support for sustainability may not translate into definitive implementation plans or timelines	Action oriented planning with defined responsibilities, staged implementation and measurable targets for water savings and reuse

Table 3 (continued)

COM-B domain	TDF domain	Barriers	Facilitators
Reinforcement		Limited feedback, incentives or recognition for services that adopt water conservation strategies	Reporting water savings, cost savings and environmental gains back to staff and leadership to reinforce behaviour change
Optimism/pessimism		Scepticism that meaningful savings can be achieved or that system-level change is realistic in complex healthcare settings	Use of local pilot projects and real-world case studies to show that implementation is feasible and can generate measurable benefit

Appendix 1 Search strategy

PubMed

("Wastewater"[Mesh] OR Water[Title/Abstract]).

AND

(Conservation[Title/Abstract] OR Reuse*[Title/Abstract] OR Save*[Title/Abstract] OR Saving[Title/Abstract] OR spar*[Title/Abstract] OR waste*[Title/Abstract] OR recycl*[Title/Abstract]).

AND

(Hemodialysis[Title/Abstract] OR hemodialysis[Title/Abstract]).

Embase

('water'/exp OR Water:ti,ab).

AND

(Conservation:ti,ab OR Reuse*:ti,ab OR Save*:ti,ab OR Saving:ti,ab OR spar*:ti,ab OR waste*:ti,ab OR recycl*:ti,ab).

AND

('hemodialysis'/exp OR Hemodialysis:ti,ab OR hemodialysis:ti,ab).

Web of Science

((TI=Water OR AB=Water)).

AND

((TI=Conservation OR AB=Conservation) OR (TI=Reuse* OR AB=Reuse*) OR (TI=Save* OR AB=Save*) OR (TI=Saving OR AB=Saving) OR (TI=spar* OR AB=spar*) OR (TI=waste* OR AB=waste*) OR (TI=recycl* OR AB=recycl*)).

AND

((TI=Haemodialysis OR AB=Hemodialysis) OR (TI=hemodialysis OR AB=hemodialysis)).

Scopus

(TITLE-ABS(Water)).

AND

(TITLE-ABS(Conservation) OR TITLE-ABS(Reuse*) OR TITLE-ABS(Save*) OR TITLE-ABS(Saving) OR TITLE-ABS(spar*) OR TITLE-ABS(waste*) OR TITLE-ABS(recycl*)).

AND

(TITLE-ABS(Hemodialysis) OR TITLE-ABS(hemodialysis)).

CINAHL

(TI Water OR AB Water)).

AND

((TI Conservation OR AB Conservation) OR (TI Reuse* OR AB Reuse*) OR (TI Save* OR AB Save*) OR (TI Saving OR AB Saving) OR (TI spar* OR AB spar*) OR (TI waste* OR AB waste*) OR (TI recycl* OR AB recycl*)).

Table 4 Summary of water conservation strategies reported outcomes and implementation considerations

What's known till date on this topic?

Haemodialysis is a water-intensive therapy with a significant environmental footprint. Water conservation efforts have focussed on reducing dialysate flow rate and reusing reverse osmosis RW, but evidence remains fragmented across clinical services and feasibility studies

What does our study add?

This scoping review integrates evidence on dialysate flow modification and RW reuse, showing that both can reduce water consumption in haemodialysis. It also highlights differences in safety, implementation, and governance, and reveals a gap between interest in sustainable dialysis and evidence from real-world implementation and evaluation

Future recommendations on this topic

Future research should focus on the implementation, scalability, and long-term impact of water conservation strategies. Priorities include standardised water-use reporting, clinical and operational evaluations, and robust economic analyses across healthcare settings. Reject water reuse studies should align with fit-for-purpose governance frameworks and address emerging contaminants such as microplastics. Implementation science may help overcome behavioural, organisational, and infrastructure barriers to wider adoption in haemodialysis

AND

((TI Hemodialysis OR AB Hemodialysis) OR (TI haemodialysis OR AB haemodialysis)).

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Artificial intelligence statement Generative artificial intelligence tools were used in a limited capacity to support the visual design and editing of selected figures and infographics. All scientific content, data, labels, and interpretations were provided, checked, and approved by the authors, who retain full responsibility for the accuracy and integrity of the final manuscript and figures.

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